

Autonomous Synthesis of Hard RTL Designs via Iterative Repair and Modular Decomposition

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Abstract

Large language models generate short RTL fragments but fail on benchmark-level hardware synthesis. ArchXBench exposes this gap: frontier models solve lower-level tasks but produce near-zero pass rates from Level 4 onward under direct prompting. This paper evaluates whether verifier-grounded autonomous synthesis can cross that failure frontier. Four conditions are compared on ArchXBench Levels 3–6: single-shot generation (C1), monolithic golden-feedback repair (C2g), decomposition-guided iterative repair (C4i), and testbench-localized decomposition (C4tl). C4i solves five Level-3 numerical kernels at 3/3 seeds where both C1 and C2g fail. C4tl solves all four core Level-4 designs (including 16-point FFT/IFFT) at 5/5 seeds. Golden-verified Level-5 and Level-6 solves are obtained on convolution, image-processing, and AES designs. Decomposition is not universally superior: C2g solves several Level-5 rows that decomposed methods do not. A second contribution is an executable-contract audit that separates original-benchmark solves from minimally repaired contract fixtures, exposing both benchmark defects and genuine unsolved capability boundaries. Code and artifacts are available.¹

CCS Concepts

• **Hardware** → **Hardware description languages and compilation**; **Electronic design automation**; • **Software and its engineering** → *Automatic programming*.

Keywords

RTL synthesis, Verilog generation, LLM agents, CEGIS, hardware verification, benchmark evaluation

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1 Introduction

Autonomous code agents close executable loops: generate a candidate, run a checker, inspect failures, and repair. For software tasks, this loop has changed what counts as a meaningful benchmark result. For hardware design, the question is harder: can an autonomous agent synthesize RTL that survives simulation, protocol assumptions, and golden-output comparison?

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RTL correctness depends on bit widths, signedness, clocking, reset protocols, pipeline latency, and streaming handshakes. A design may compile and still be functionally wrong. A simulator may run to completion and still produce incorrect golden traces. Hard RTL synthesis therefore tests whether autonomous agents can handle low-level, state-dependent, timing-sensitive design, not just surface-level code generation.

This paper studies the problem on ArchXBench, a benchmark suite for complex RTL synthesis [8]. Unlike VerilogEval [4] and RTLLM [5], which target smaller designs, ArchXBench spans arithmetic kernels, FFTs, filters, image processing, matrix multiplication, and cryptography across six difficulty levels. The ArchXBench paper reports that frontier models solve lower levels but fail from Level 4 onward under zero-shot pass@5. This Level-4 boundary is the starting point of the present study. If an autonomous synthesis loop can solve rows beyond it, the evaluation question changes from “can an LLM emit a complete design in one shot?” to “can a verifier-grounded agent converge to correct RTL?”

The central question is: *can verifier-grounded autonomous synthesis solve hard RTL benchmark tasks that direct generation does not solve, while maintaining a clear distinction between model failures and benchmark-contract failures?* The answer is partially affirmative. C4i solves five Level-3 numerical kernels at 3/3 seeds where C1 and C2g both fail. C4tl solves all four core Level-4 designs at 5/5 seeds, including the 16-point FFT and IFFT. Golden-verified solves extend to Level-5 convolution and image processing and to Level-6 AES encryption and decryption. At the same time, C2g is a strong baseline: it solves several Level-5 rows and multiple repaired-contract designs that decomposed methods do not. The contribution is not a claim that decomposition dominates monolithic repair. It is evidence that verifier-grounded synthesis, in both monolithic and decomposed forms, can cross parts of the reported hard-task frontier, and that benchmark contracts must be audited before either successes or failures are interpreted.

This paper makes three contributions:

- (1) It demonstrates verifier-grounded autonomous synthesis solving hard ArchXBench rows beyond the direct-generation frontier, including robust Level-4 original-contract solves and golden-verified Level-5/Level-6 solves.
- (2) It compares four synthesis conditions under a common artifact discipline, showing where each succeeds or fails.
- (3) It introduces an executable-contract audit that separates original-contract results, repaired-contract results, and held rows.

2 Related Work

2.1 LLM-Based RTL Generation

VerilogEval introduced an HDLBits-style benchmark for Verilog generation with automatic functional checking [4]. RTLLM proposed an open benchmark for design RTL generation with large language models [5]. Both benchmarks made RTL generation measurable, but most tasks are small instructional circuits rather than deeply structured systems.

More recent work moves beyond direct prompting. AutoChip uses compiler and simulator feedback to improve HDL generation [10]. VerilogCoder introduces graph-based planning and AST-based waveform tracing [1]. AutoVeriFix uses simulation discrepancies to correct LLM-generated Verilog [9]. Spec2RTL-Agent studies LLM-agent systems for specification-to-RTL generation [14]. VFlow searches over agentic workflows using Monte Carlo tree search [13]. HDLxGraph uses HDL-specific graph retrieval to support repository-level HDL tasks [15]. Evolutionary methods (REvolution, Evolve, CO-EVO) use inference-time search or co-evolutionary optimization for RTL correctness and power/performance/area [2, 6, 7]. RTL-BenchMT studies agent-assisted benchmark maintenance, including flawed-case revision and overfitting detection [12]. FormalRTL integrates software references and formal equivalence checking into a multi-agent RTL synthesis flow [3]. RTL-BenchLS proposes a large formally verified RTL benchmark for reasoning and generation tasks [11].

This study differs in emphasis. It evaluates on ArchXBench Levels 3–6, including rows at and above the reported Level-4 frontier. It separates original benchmark contracts from repaired executable contracts. This distinction matters because an RTL benchmark can fail either because the generated design is wrong or because the released testbench, file format, or golden comparator is inconsistent. RTL-BenchMT validates that benchmark ambiguity is a first-order issue; FormalRTL and RTL-BenchLS show the value of stronger executable or formal oracles. The present study uses benchmark-contract audit to support a hard-ArchXBench synthesis study rather than to propose a general benchmark-maintenance system.

Table 1 summarizes the positioning.

2.2 Verifier-Grounded Synthesis

Counterexample-guided inductive synthesis (CEGIS) uses a verifier to reject incorrect candidates and guide the next attempt. LLM-based repair loops instantiate a related pattern: generate code, run tests, inspect errors, and repair. For RTL, verifier feedback is heterogeneous: failures arise from syntax, elaboration, simulation, timing, protocol mismatch, file-output mismatch, or incorrect golden references. The present pipeline treats simulator and golden-output feedback as first-class signals and audits the benchmark contract when executable artifacts disagree.

3 Problem Setting

Each ArchXBench design provides a natural-language specification and executable assets: Verilog testbenches, input

files, output files, and comparison scripts. An autonomous synthesis condition receives the task and emits RTL. For self-checking designs, a run succeeds only when the official testbench passes all checks. For file-output designs, simulation completion is not sufficient; the generated output must match the golden output element by element.

Three evaluation classes are distinguished:

- **Original-contract results:** runs against the released benchmark as-is.
- **Repaired-contract results:** runs against a copied fixture with minimally repaired infrastructure inconsistencies.
- **Held or excluded rows:** tasks where the released contract remains too ambiguous for a credible positive result.

Table 2 gives the contract taxonomy used during the audit.

4 Method

4.1 Evaluated Conditions

Table 3 summarizes the operational differences among the four conditions. All conditions receive the same specification, module interface, and released testbench. The difference is how much executable feedback enters the loop and whether the design is treated as a monolith or decomposed into modules.

C1: direct generation. The model receives the specification and testbench excerpt and emits RTL five independent times. The candidate with the highest pass count under the executable checker is retained. No repair feedback is used.

C2g: monolithic golden-feedback repair. The model studies the specification, interface, and testbench, then emits a complete top module. Each round compiles and simulates the candidate. Compile errors, self-checking test failures, or golden-output mismatches (with index-level expected-versus-actual values) are appended to the conversation. The model returns a corrected module. The conversation is pruned to the first two and last sixteen messages when it exceeds twenty entries. The loop terminates when all checks pass or after 30 rounds. This is a deliberately strong baseline: it uses the same executable oracle as decomposed methods without paying module-integration risk.

C4i: decomposition-guided iterative repair. C4i asks the model to decompose the task into 3–7 submodules with explicit interfaces, a top-level wrapper, and reference implementations. Submodules are constrained to be combinational where possible; sequential logic is concentrated in the top module. The reference composition (top plus all reference submodules) is simulated against the system testbench as a sanity check.

After decomposition, each submodule is solved in sequence. The model receives the submodule specification, its position in the pipeline relative to neighbors, the reference implementation, and a testbench excerpt. A study prompt requires the model to explain its understanding before writing code. During repair, all unsolved submodules are held at their reference implementations, isolating bugs to the module under repair. Each submodule receives $\lfloor 28/n \rfloor$ repair rounds for n

Table 1: Positioning relative to recent LLM-aided RTL work. The key distinction is a hard-ArchXBench study combining verifier-grounded synthesis with explicit original-contract versus repaired-contract accounting.

Work	Primary target	Feedback / search	Benchmark style	Difference from this study
VerilogEval / RTLLM	Generation benchmarks	Testbench eval.	HDLBits / moderate RTL	Establishes automatic evaluation; mostly easier than ArchXBench L4–L6
AutoChip / AutoVeriFix	HDL repair	Compiler, simulator, oracle	Repair/generation tasks	Iterative correction is related; no benchmark-contract audit
VerilogCoder / Spec2RTL	Autonomous agents	Planning, tracing, multi-agent	VerilogEval / RTLLM	Strong agents; not focused on ArchXBench hard-level contracts
VFlow	Workflow search	MCTS over workflows	Generation tasks	Searches workflows; does not study contract validity
REvolution / EvoIVE / COEVO	Evolutionary search	Population search, PPA	VerilogEval / RTLLM	Search-oriented; does not study benchmark contracts
RTL-BenchMT	Benchmark maintenance	Agent-assisted revision	VerilogEval / RTLLM / CVDP	Validates contract concerns; focused on maintenance
FormalRTL / RTL-BenchLS	Formal verification	Equivalence checking	Formally verified suites	Stronger oracles; not focused on ArchXBench hard rows
This work	Hard RTL synthesis	Golden repair, decomposition, audit	ArchXBench L3–L6	Separates original, repaired, and held rows

Table 2: Executable-contract taxonomy. Each category changes how a run is interpreted.

Contract type	Paper treatment
Self-checking TB	Pass all official checks; native score is sufficient.
File-output w/ golden	Post-simulation golden comparison required; simulation completion is diagnostic only.
Display-only checker	Not a solve until expected values are converted into executable checks.
Stale comparator	Repair in a separate fixture; report separately.
Spec/TB disagreement	Exclude unless a principled source of truth exists.
Missing oracle	Hold; do not claim positive results.

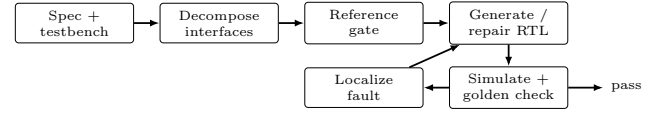
Table 3: Operational differences among synthesis conditions. “Ref. gate” indicates whether the decomposition’s reference composition must pass the system testbench before synthesis proceeds. “Local repair” describes the granularity of repair targeting.

Cond.	Decomp.	Ref. gate	Golden fb.	Local repair	Budget
C1	no	–	no	–	5 shots
C2g	no	–	yes	whole design	30 rounds
C4i	yes	sanity	yes	per module	≈28 rds
C4tl	yes	strict	yes	culprit only	3+≈28 rds

submodules. Golden-output mismatches are fed back with specific failing indices and expected values.

C4tl: testbench-localized decomposed repair. C4tl uses the same decomposition structure as C4i but adds two mechanisms: a strict reference gate and trace-lifted fault localization.

The reference gate requires the composed reference implementation to pass the system-level testbench before synthesis begins. If the reference fails, the decomposition is retried up to three times with failure feedback augmenting the specification. If no valid decomposition is found, the run aborts. This prevents ungrounded decompositions from entering the synthesis loop.

**Figure 1: C4tl pipeline. The reference gate validates the decomposition. On simulation failure, trace-lifted localization identifies the culprit module. Only that module is repaired.**

After initial candidate generation for all submodules, C4tl runs the full composed design against the system testbench. On failure, trace-lifted localization identifies the culprit module: each candidate submodule is swapped with its validated reference one at a time while all other candidates remain fixed. The module whose swap produces the largest score improvement is selected for repair. Only that module is repaired using the failing simulator output, golden mismatch details, per-module swap scores, and the reference implementation. The model is instructed to trace the datapath for the first failing input, compute expected intermediate values, identify the root cause, and then fix. This loop repeats until the system checker passes or the budget is exhausted.

The localization mechanism is the central algorithmic distinction. C4i repairs modules in a fixed order; C4tl uses system-level evidence to identify which module is most implicated. For tightly coupled pipelines such as the 16-point FFT, this targeted approach avoids diffuse repair of modules that may not be at fault.

4.2 Artifact and Verification Discipline

Each run is recorded under a repository artifact tree with a structured result file containing the design, condition, model, seed, pass counts, golden counts, LLM-call count, wall time, and generated RTL. File-output designs require strict golden verification: the generated output is compared element by element against the golden reference with a tolerance of ± 1 on integer values. Native simulation completion alone is treated as diagnostic evidence, not as a solve.

Table 4: Aggregate profile across all Level-3–6 GPT-5.5 artifact rows, including original-contract, repaired-contract, and diagnostic runs. Because the aggregate is heterogeneous, this table is descriptive rather than a controlled per-design comparison.

	Cond.	Runs	Clean	Avg. calls	Avg. wall
C1		100	16	5.0	9.2 min
C2g		69	41	15.5	22.6 min
C4i		125	76	14.6	14.8 min
C4tl		143	83	7.4	9.3 min

5 Experimental Setup

5.1 Benchmark Scope

The evaluation covers ArchXBench Levels 3–6: iterative arithmetic, FFT/IFFT, floating-point pipelines, FIR filters, image-processing kernels, matrix multiplication, and AES. Lower levels are excluded because the ArchXBench paper reports that current models already solve them.

5.2 Seeds, Model, and Metrics

Main tables use seeds 42, 123, and 456. Two additional seeds (789, 1024) provide robustness evidence for C4tl Level-4 rows. These are fixed seeds chosen for reproducibility, not tuned seeds. The primary model is GPT-5.5 as recorded in the repository artifacts.

A clean solve requires all official checks to pass (self-checking designs) or all golden outputs to match (file-output designs). Solve rates are reported over seeds.

5.3 Implementation and Cost

All methods run as executable synthesis loops. C1 uses five independent generations. C2g uses a 30-round monolithic repair loop. C4i allocates a per-module budget from a 28-round total. C4tl uses up to three decomposition attempts followed by a bounded localization-and-repair loop. All simulations use Icarus Verilog.

Table 4 reports aggregate statistics across all repository-local GPT-5.5 Level-3–6 artifact rows. C2g uses fewer total runs but more calls per run. C4tl averages 7.4 calls per run, roughly half of C4i, because targeted repair avoids diffuse iteration.

Table 5 reports per-design statistics for selected solved rows, averaged over seeds. These illustrate the practical cost of individual solves: the Level-4 FFT solves in roughly 5 minutes and 6 LLM calls, while harder designs such as `harris_corner_detection` require 10.7 calls and 9.4 minutes.

6 Crossing the Original-Contract Frontier

Table 6 reports the main original-contract results.

Level 3. C4i provides the strongest method-value evidence: it solves five numerical kernels at 3/3 seeds where C1 scores 0/3–0/5 and C2g scores 0/3–1/5. C4tl matches C4i on four of five rows but solves `gauss_siedel` at only 1/3. For context, C4i decomposes `gauss_siedel` into three submodules (reciprocal unit, iteration step, solution selector), allowing

Table 5: Per-design cost for selected solved rows (seed-averaged). Level-4 rows average over 5 seeds; all others over 3.

Design	Method	Avg. calls	Avg. wall
<code>fp_multiplier</code> (L3)	C4i	5.0	5.2 min
<code>gauss_siedel</code> (L3)	C4i	14.3	11.9 min
<code>fft_16pt_iterative</code> (L4)	C4tl	6.0	5.1 min
<code>ifft_16pt_iterative</code> (L4)	C4tl	6.2	5.5 min
<code>harris_corner</code> (L5)	C4i	10.7	9.4 min
<code>aes_encryption</code> (L6)	C4i	7.3	7.1 min

each to be solved and tested in isolation. The negative result on `newton_raphson_polynomial` is an original-checker ceiling: three of its 100 assertions are structurally unsatisfiable, capping the achievable score at 97/100.

Level 4. C4tl solves all four core Level-4 rows at 3/3 main seeds and 2/2 robustness seeds, for 5/5 total. The 16-point FFT and IFFT rows are the strongest hard-frontier evidence because they are self-checking original-contract solves with no benchmark repair. C4tl decomposes `fft_16pt_iterative` into four submodules (bit-reverse mapper, twiddle ROM, butterfly unit, output scaler) and uses trace-lifted localization to identify which module is at fault after each simulation failure. C2g also solves the FFT/IFFT rows at 3/3; C4i does not. One explanation is that C4i’s fixed per-module repair order is less effective than C4tl’s targeted localization on tightly coupled pipeline designs. The pipeline rows (`fp_adder_pipeline`, `fp_mult_pipeline`) are solved by all conditions.

Levels 5–6. Golden-verified solves are obtained on `conv1d` (16/16), `harris_corner_detection` (16,384/16,384), `aes_encryption` (8/8), and `aes_decryption` (8/8), all at 3/3 seeds under C4i. C4i decomposes AES encryption into six submodules (S-box, SubBytes, ShiftRows, MixColumns, round function, key expansion), each implemented and tested against the system testbench with other modules held at their references. Several Level-5 rows are solved only by C2g: `conv2d` (4,096/4,096), `dct_idct_8pt_pipeline` (16/16), and `unsharp_mask` (65,536/65,536). These rows demonstrate that monolithic repair is the stronger strategy when the design is compact enough for whole-design feedback or when the primary difficulty is a global convention such as file format or fixed-point encoding.

The results are not presented as a universal decomposition win. C2g is a serious baseline: it matches or outperforms decomposition on 8 of 17 original-contract rows. The most accurate interpretation is that decomposition provides value on specific hard rows and gives a structured synthesis process, while monolithic golden-feedback repair is often sufficient and sometimes better.

7 Executable-Contract Audit

Hard RTL benchmarks are useful only when their executable contracts reflect the intended task. During evaluation, several rows were found to contain inconsistent specifications, stale comparators, display-only checkers, or output-schema

Table 6: Original-contract results on ArchXBench. Values are clean solves over seeds; bold 3/3 indicates 3/3 on the main seed set. GV: golden-verified file-output row. “+2” means seeds 789 and 1024 also solve, giving 5/5 total.

Level	Design	C1	C2g	C4i	C4tl	Evidence
L3	fp.adder	0/5	1/5	3/3	3/3	decomposition wins
L3	fp.multiplier	0/5	0/3	3/3	3/3	decomposition wins
L3	gauss.siedel	0/3	0/3	3/3	1/3	C4i strongest
L3	gradient.descent	0/3	0/3	3/3	3/3	decomposition wins
L3	newton_raphson_sqrt	0/3	0/3	3/3	3/3	decomposition wins
L3	newton_raphson_poly.	0/3	0/3	0/3	0/3	checker ceiling at 97/100
L4	fft.16pt.iterative	0/3	3/3	0/3	3/3 (+2)	hard-row solve
L4	ifft.16pt.iterative	0/3	3/3	0/3	3/3 (+2)	hard-row solve
L4	fp.adder.pipeline	3/3	3/3	3/3	3/3 (+2)	all methods solve
L4	fp.mult.pipeline	3/3	3/3	3/3	3/3 (+2)	all methods solve
L5	conv1d	3/3	3/3	3/3	1/3	GV
L5	conv2d	0/3	3/3	0/3	0/3	GV; C2g strongest
L5	dct.idct.8pt.pipelined	0/3	3/3	0/3	0/3	GV; C2g strongest
L5	harris.corner.detection	0/3	3/3	3/3	0/3	GV
L5	unsharp_mask	0/3	3/3	0/3	0/3	GV; C2g strongest
L6	aes.encryption	3/3	3/3	3/3	2/3	GV
L6	aes.decryption	1/3	3/3	3/3	1/3	GV; repair improves C1

Table 7: Repaired-contract results, separated from original ArchXBench claims. These distinguish benchmark-contract failures from model capability failures.

Design	C2g	C4i	C4tl
conv_3d	3/3	2/3	0/3
multich_conv2d	3/3	3/3	3/3
quantized_matmul	3/3	3/3	0/3
fp.band.pass.fir	3/3	0/1	0/1
fp.high.pass.fir	3/3	0/1	0/1
newton_raphson.poly.	3/3	1/3	1/3
systolic_gemm	0/3	0/3	0/3

mismatches. The released benchmark is not overwritten. Repaired fixtures are kept under a separate artifact root and reported separately.

The audit follows a fixed protocol. First, the released fixture is run unchanged. Second, the checker is classified (self-checking, file-output with golden, display-only, stale, specification-inconsistent, or missing-oracle). Third, file-output rows are judged only by post-simulation golden comparison. Fourth, infrastructure repairs are applied only in copied fixtures and only when the intended oracle is recoverable from released assets. Fifth, unresolved rows are held rather than converted into positive results.

Table 7 reports the repaired-contract evidence. These rows are not counted as original ArchXBench solves.

The audit produces both positive and negative evidence. Repairing the executable contract for `multich_conv2d` and `quantized_matmul` makes the intended task measurable and solvable. By contrast, converting the display-only `systolic_gemm` checker into executable checks does not produce a win: all methods remain at 0/3. This is a genuine remaining capability boundary, not a benchmark-repair success.

Several rows remain held or excluded. The L4 FIR family is excluded because the specification and testbench disagree on filter coefficients. The L6 `fp_low_pass_fir` is held because

the released files do not expose an explicit coefficient oracle. The L6 `fft_streaming_64pt` is excluded because the released input/output contract has unresolved schema and numeric-encoding ambiguities.

8 Discussion

8.1 When Decomposition Helps and When It Does Not

Decomposition is most effective when a design factors naturally into modules and when failure localization reduces the repair burden. The Level-3 numerical kernels and Level-4 FFT/IFFT rows illustrate this: C4i and C4tl solve rows that both C1 and C2g fail on. C4i fails on the FFT/IFFT rows where C4tl succeeds. One explanation is that C4i’s fixed per-module repair order is less effective for tightly coupled pipeline stages; C4tl’s trace-lifted localization identifies the culprit module and avoids diffuse iteration on modules that are not at fault.

By contrast, C2g solves several Level-5 rows that neither C4i nor C4tl can solve. This happens when the design is compact enough for whole-design repair, when module boundaries introduce integration risk, or when the primary difficulty is a global convention such as file format or fixed-point encoding. The paper’s claim is therefore conservative: decomposition is a powerful synthesis strategy for some hard rows, but monolithic repair remains strong and should be treated as a primary baseline.

8.2 Why Benchmark Contracts Matter

LLM-aided RTL synthesis papers increasingly rely on executable benchmarks. If the executable contract is ambiguous, a model can fail for the wrong reason or pass the wrong task. The issues observed in this study include hidden testbench assumptions, stale comparators, file-output schema mismatches, display-only checkers, unsatisfiable residual checks, and missing oracle information. Separating original-contract

and repaired-contract results improves the reliability of future comparisons on hard RTL benchmarks.

9 Limitations

The evaluated methods do not establish universal dominance over monolithic repair; C2g should be treated as a primary baseline. Some historical rows are trusted score-only rather than artifact-backed and are labeled accordingly. Repaired-contract evaluation depends on judgment about what constitutes an infrastructure repair rather than a semantic benchmark change; this risk is mitigated by keeping repaired fixtures separate and excluding unresolved rows. The empirical study is concentrated on RTL synthesis; transfer to other executable synthesis domains is not established here.

10 Conclusion

This study evaluated autonomous RTL synthesis on hard ArchXBench designs using iterative repair, modular decomposition, and strict simulator/golden feedback. Original-contract solves span Levels 3–6, including robust Level-4 FFT/IFFT rows and golden-verified Level-5/Level-6 designs. Monolithic golden-feedback repair is a strong baseline. Decomposition is powerful but not universally superior. Benchmark-contract defects can obscure both progress and failure.

Hard autonomous RTL synthesis requires structured generation, executable feedback, and trustworthy benchmark contracts together. Without all three, both successes and failures can be misinterpreted.

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